Tim Book

An Intro to Recommender Systems

GENERAL ASSEMBLY

Agenda

- What is a recommender system?
- What kinds of recommender systems are there?
- Cosine similarity
- Python!

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What is a recommender system?

Simply put, a **recommender system** is a system that can help recommend new items in a service to someone. You probably interact with many of them per day:

- Amazon: “You might also like…”
- Netflix: “Recommended for Tim…”
- Goodreads: “Because you liked X, you might like Y.”
- Spotify: “Tim’s Daily Mix”

The Netflix Prize

In 2006, little-known DVD rental service **Netflix** issued a challenge: to beat their recommender system and win $1,000,000! A winner was declared in 2009. Netflix ended up not using their algorithm in the end.

You can read more about it here!



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Types of Recommenders

Different kinds of recommenders (more on each soon)

1. Collaborative filtering
2. Content-based
3. Knowledge-based
4. Hybrid/Ensemble-based

Collaborative Filtering

Collaborative filtering recommends items via **other users' ratings**.

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Option 1: User-based.

You're User A. You rate some movies.

User B has rated movies. User B's preferences look a lot like your preferences.

*Our recommendation: You should watch this movie that User B liked, because **you two are similar users**.*

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*Our recommendation: You should watch this movie that User B liked, because **you two are similar users**.*

Option 2: Item-based.

You're User A. You rate Movie 1 well.

The people who rate Movie 1 well also tended to rate Movie 2 well.

*Our recommendation: You should watch Movie 2, because you like Movie 1, and **Movies 1 and 2 are similar movies (by ratings)**.*

Case Study:

Collaborative filtering the MovieLens dataset with Cosine Similarity

The MovieLens Data

Movie	User	Rating
Movie 1	User 1	5
Movie 1	User 2	4
Movie 1	User 3	1
Movie 2	User 1	1
Movie 2	User 2	1
Movie 2	User 3	5
Movie 3	User 1	4
Movie 3	User 2	5
Movie 3	User 3	1

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Movie 1	User 1	5
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Movie 2	User 2	1
Movie 2	User 3	5
Movie 3	User 1	4
Movie 3	User 2	5
Movie 3	User 3	1



User-Based Setup

	Movie 1	Movie 2	Movie 3
User 1	5	1	4
User 2	4	1	5
User 3	1	5	1

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The MovieLens Data

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Movie 1	User 1	5
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User-Based Setup

	Movie 1	Movie 2	Movie 3
User 1	5	1	4
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Item-Based Setup

	User 1	User 2	User 3
Movie 1	5	4	1
Movie 2	1	1	5
Movie 3	4	5	1

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Collaborative filtering: User-based

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **users** to be the object of study.

	Movie 1	Movie 2	Movie 3
User 1	5	1	4
User 2	4	1	5
User 3	1	5	1

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Collaborative filtering: User-based

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **users** to be the object of study.

Who is most similar to User 1?

	Movie 1	Movie 2	Movie 3
User 1	5	1	4
User 2	4	1	5
User 3	1	5	1

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Collaborative filtering: Item-based

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **movies** to be the object of study.

	User 1	User 2	User 3
Movie 1	5	4	1
Movie 2	1	1	5
Movie 3	4	5	1

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Collaborative filtering: Item-based

Remember in data science, we prefer

1 row = 1 observation

So in this view, we're considering **movies** to be the object of study.

If I like Movie 1, what other movie might I like?

	User 1	User 2	User 3
Movie 1	5	4	1
Movie 2	1	1	5
Movie 3	4	5	1

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Similarity Metric of Vectors

So, creating a movie recommender is as simple as measuring the **similarity of every movie based on its ratings!** That is, the “similarity” of the movie vectors.

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We’ll learn something new for this, but any ideas based on stuff we already know how to do?

Similarity Example

	Mamma Mia! Here We Go Again	*Frozen*
Tim	4	1
Jamie	5	2
Heather	1	5

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Similarity Example

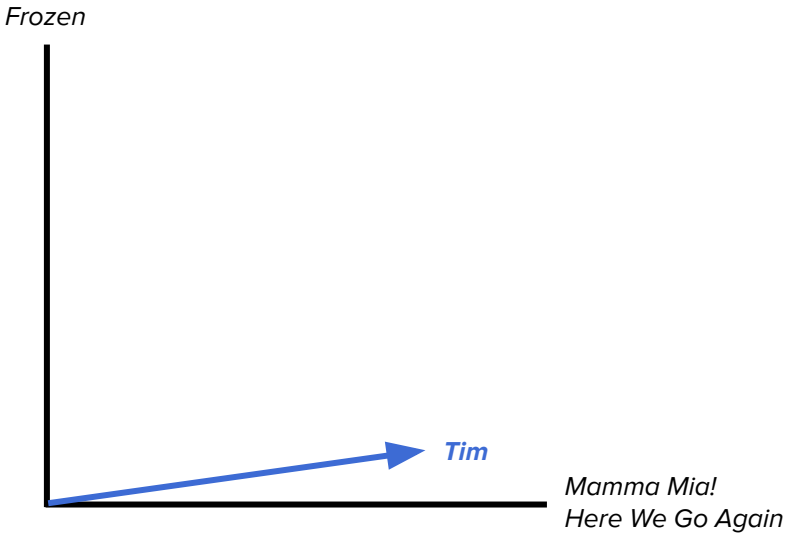
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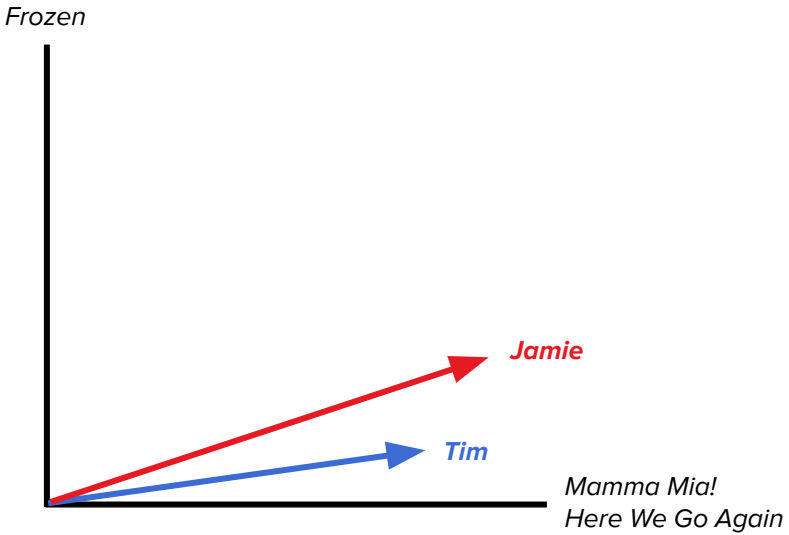
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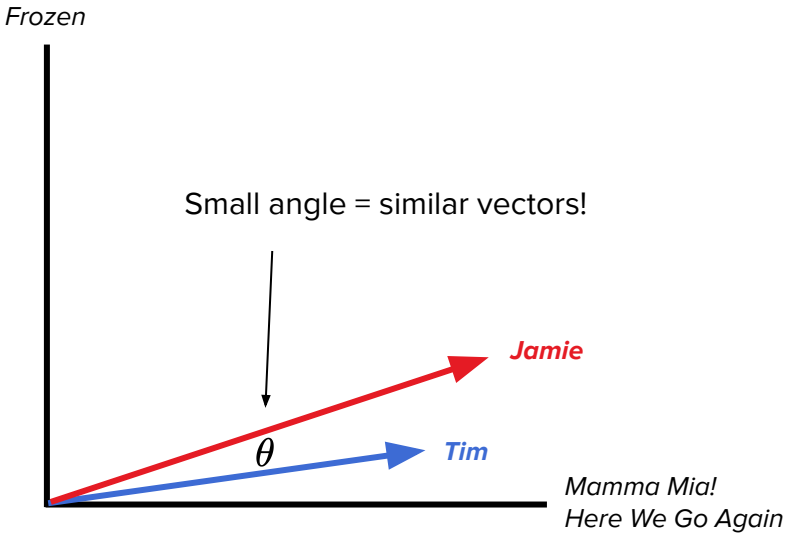
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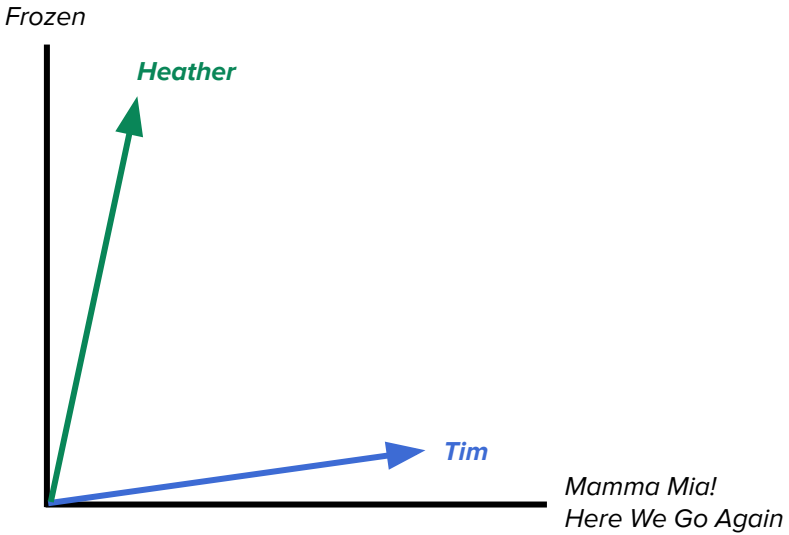
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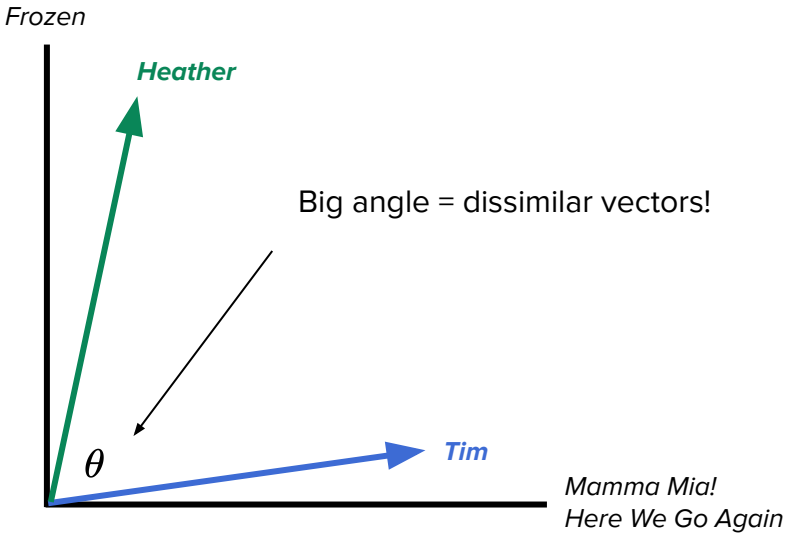
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Similarity Example

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Cosine Similarity

$$\text{cosine sim} = \cos\theta = \frac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

Trig reminder:

- cos 0° = 1
- cos 90° = 0

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Tim vs Jamie Example:

	x	*y*
tim	4	1
jamie	5	2
heather	1	5

$$
\begin{aligned}
\text{similarity} &= \frac{\mathbf{t} \cdot \mathbf{j}}{\|\mathbf{t}\| \|\mathbf{j}\|} \\
&= \frac{4 \times 5 + 1 \times 2}{\sqrt{4^2 + 1^2}\sqrt{5^2 + 2^2}} \\
&= \frac{22}{\sqrt{17}\sqrt{29}} \\
&= 0.991
\end{aligned}
$$

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Tim vs Heather Example:

	x	*y*
tim	4	1
jamie	5	2
heather	1	5

$$
\begin{aligned}
\text{similarity} &= \frac{\mathbf{t} \cdot \mathbf{h}}{\|\mathbf{t}\|\|\mathbf{h}\|} \\
&= \frac{4 \times 1 + 1 \times 5}{\sqrt{4^2 + 1^2}\sqrt{1^2 + 5^2}} \\
&= \frac{9}{\sqrt{17}\sqrt{26}} \\
&= 0.428
\end{aligned}
$$

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Let’s see for ourselves!



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Types of Ratings

Explicit Rating Systems

Many rating systems are **explicit**, as in, you're rating the item on purpose. For example, many are **interval-based**:

Leave a review for this book:

Hated it!	Disliked it.	It was ok.	Liked it!	Loved it!
★☆☆☆☆	★★☆☆☆	★★★☆☆	★★★★☆	★★★★★

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Explicit Rating Systems

Many rating systems are **explicit**, as in, you're rating the item on purpose. For example, many are **interval-based**:

Leave a review for this book:

Hated it!	Disliked it.	It was ok.	Liked it!	Loved it!
-2	-1	0	1	2

Explicit Rating Systems

Explicit ratings can be **binary**.

Did you like this book?

Dislike	Like
☐	☐
0	1

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Implicit Rating Systems

Very often, ratings can be **unary**, in which case we might consider them **implicit**.

Did you like this book?

No feedback	Like
-	□
-	1

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Other Types of Recommenders

Different kinds of recommenders

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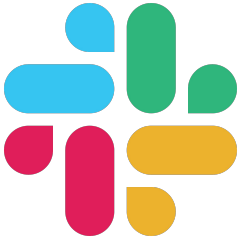
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Content-Based Recommender Systems

In **content-based** systems, it's the actual descriptive attributes of the item itself that are used to make recommendations. Consider this example taken from *Recommender Systems,* by Charu C. Aggarwal:

Consider a situation where John has rated the movie Terminator highly, but we do not have access to the ratings of other users. Therefore, collaborative filtering methods are ruled out. However, the item description of Terminator contains similar genre keywords as other science fiction movies, such as Alien and Predator. In such cases, these movies can be recommended to John.

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Using the skills you have right now, do you think you could make some kind of **content-based recommender system?**

Content-Based Recommender Systems

Sure! Probably! You could take the item's information and **cluster** it. Items in the same cluster would be more similar to each other and thus potential recommendations.

Even if you have text-data, you could use an NLP clustering technique such as **Latent Dirichlet Allocation (LDA)** which we don't learn about in our class, but is a popular technique for student capstones!

Knowledge-Based Recommender Systems

Knowledge-based systems use user-specifications to retrieve the best recommendation. A clear example of this is a website like **Yelp**:

I want food that...

- Offers dine-in
- Medium-price ($$)
- Good for kids
- Within 10 miles of my home
- Serves alcohol

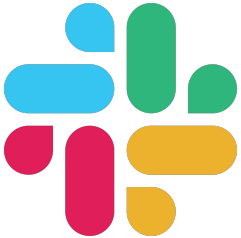
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Knowledge-Based Recommender Systems

Knowledge-based models are often used when either

1. The item being recommended is purchased by people rarely (cars, houses, etc) or
2. You don’t have any starting data with user ratings (the “cold start” problem)

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How do you think you might carry out a **knowledge-based** recommender system?

Not everything needs to be complicated...

```sql
SELECT
    restaurant_name,
    restaurant_address
FROM
    restaurants
WHERE
    offers_dine_in
    AND cost = '$$'
    AND good_for_kids
    AND distance <= 10
    AND serves_alcohol
```



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Hybrid or Ensemble-Based Systems

Few recommenders only rely on one system. It's common to combine them. For example, our quintessential knowledge-based recommender Yelp also uses user reviews to create a collaborative filtering system.

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